For the week of October 06-October 10

1) (a) Using horizontal slices, set up an integral calculating the area of the following shapes:

   i. An equilateral triangle of side length 6.

   ii. A semicircle of radius 5.

   (b) Evaluate each of the previous integrals. You may need to use a trig subtitution $x = \tan(u)$ to evaluate the second integral.

2) In this problem we will calculate the volume of a triangular pyramid of height 6, assuming the base is an equilateral triangle of side length 6.

   (a) First, determine the area of a horizontal slice of height Δh.

   (b) Set up an integral which calculates the volume of the pyramid.

   (c) Evaluate the integral.

3) Consider the region bounded by the functions $f(x) = \sqrt{x}$ and $g(x) = x$. Calculate its area using

   (a) Vertical slices (i.e. how we have calculated such areas in the past)

   (b) Horizontal slices.

4) Consider a cone of height h with base radius r.

   (a) Calculate the area of a thin slice of width Δh.

   (b) Set up an integral calculating the volume of the cone.

   (c) Evaluate the integral.

5) Find the volume of a sphere of a solid sphere r using slices.

6) Consider a shape of height 2 which at height h has horizontal slice a square of side length $1 + x^2$. Set up an integral calculating the volume of this shape and evaluate it.

7) In this problem, the integral represents the area of either a triangle or part of a circle, and the variable of integration measures a distance. Say which shape is represented, and give the base and height of the triangle or the radius of the circle. Make a sketch to support your answer showing the variable and all other relevant quantities.

   (a) $\int_0^1 3x dx$

   (b) $\int_{-9}^{9} \sqrt{81 - x^2} dx$

   (c) $\int_0^{\sqrt{15}} \sqrt{15 - h^2} dh$

   (d) $\int_0^7 5 \left(1 - \frac{h}{7}\right) dh$

The following problems are for practice only and will not appear on quizzes.

I) Textbook section 8.1, problems # 1-13, 15-23, 37-41, 52-55